

Containerized CI/CD Orchestration Platform

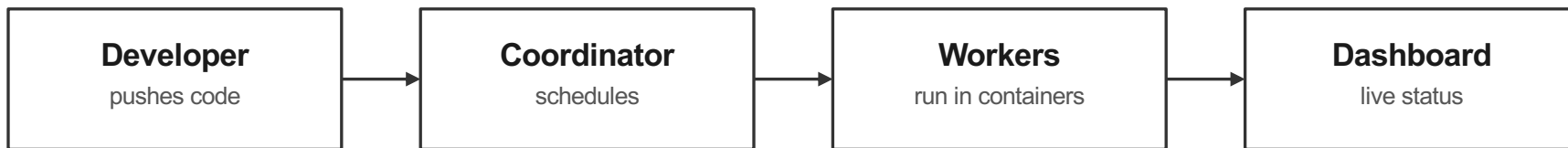
Mid-Term Project Defense



What is the project?

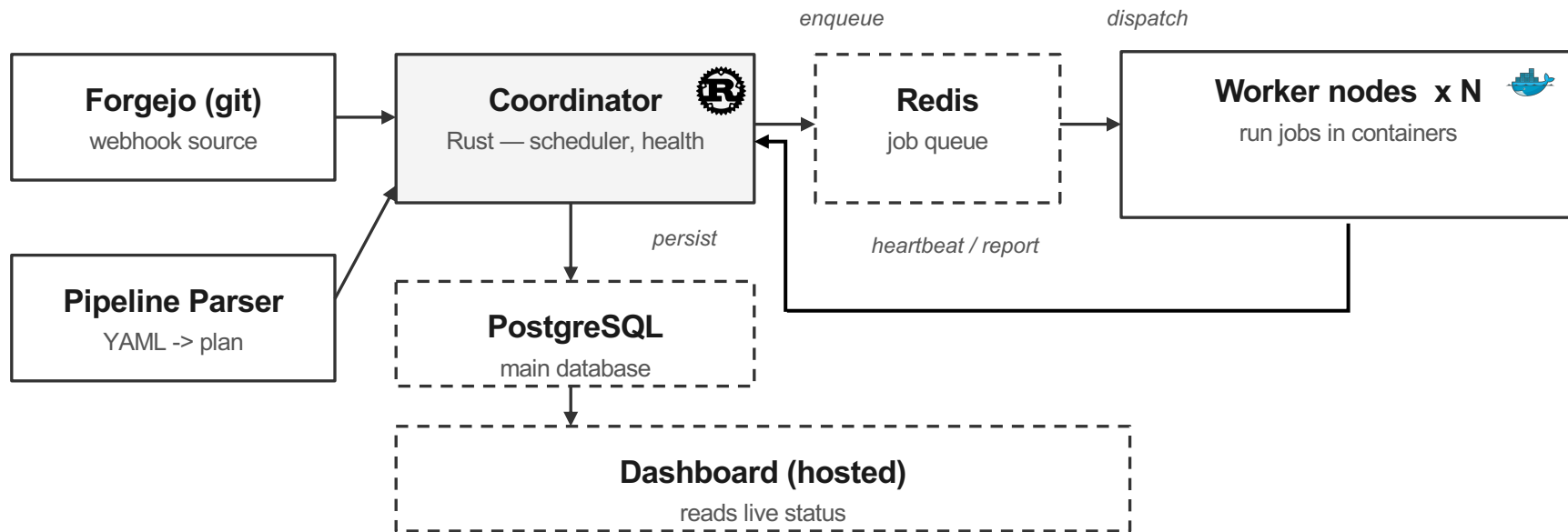
A self-hosted platform that runs build, test and deploy jobs **across many machines**, each job isolated in a container.

Unlike a single centralized server, if one worker fails its job is re-assigned automatically.



System architecture

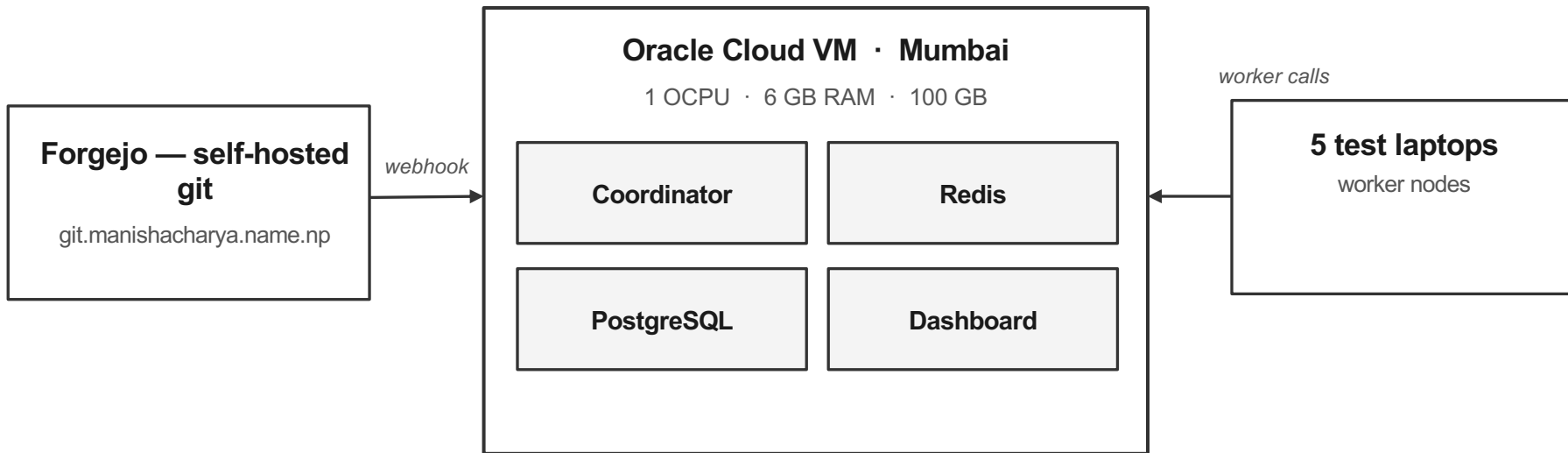
Webhook -> coordinator -> Redis queue -> workers; PostgreSQL stores everything



Solid = working now · Dashed = planned / in progress

Deployment & infrastructure

Where the platform runs



- Coordinator, Redis, PostgreSQL and the dashboard are hosted together on the Mumbai VM.
- Five physical laptops act as distributed worker nodes; Forgejo triggers pipelines via webhook.

Four modules, one system

Separate programs that talk over HTTP + JSON



Coordinator Rust

Schedules, tracks health, dispatches jobs.



Pipeline Parser Python

Reads pipeline.yml, validates, orders jobs.



Command Executor FastAPI

Runs a command in a container, reports result.



Dashboard HTML / JS

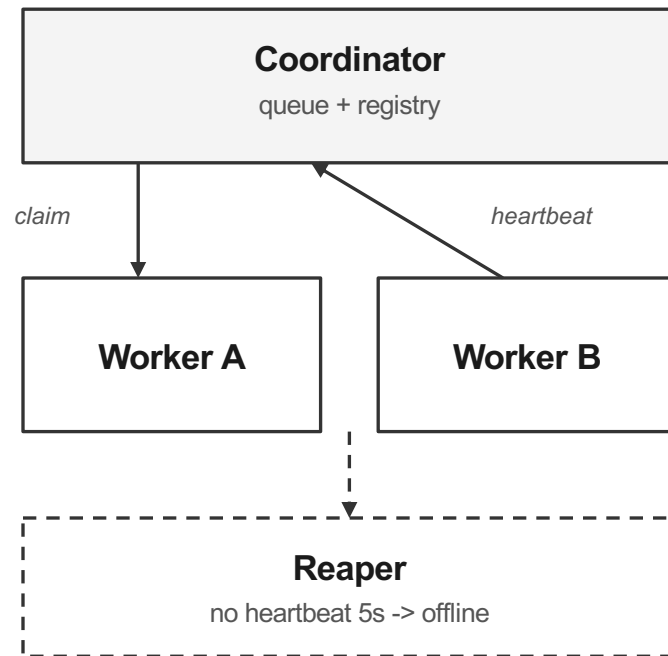
Shows pipelines, jobs and logs, live.

Done — Coordinator (the core)



The distributed engine works

- REST API: register, heartbeat, trigger, claim, report
- Job queue + worker registry in memory
- Heartbeat failure detection — a dead worker is caught automatically



cargo run -- coordinator --port 8080

SCROLL: 0/48

```
manish-ach@macs in repo: CI-CD-orchestrator on ʘ main [!+] is ʘ v0.1.0 via ʘ v1.90.0 t
ook 0s
└─λ cargo run -- coordinator --port 8080
    Finished `dev` profile [unoptimized + debuginfo] target(s) in 0.05s
    Running `target/debug/orchestrator coordinator --port 8080`
Worker recheck registered successfully!
Worker recheck registered successfully!
```

cargo run -- worker --name recheck

SCROLL: 0/23

```
manish-ach@macs in repo: CI-CD-orchestrator on ʘ main [!+] is ʘ v0.1.0 via ʘ v1.90.0 t
ook 0s
└─λ cargo run -- worker --name recheck
    Finished `dev` profile [unoptimized + debuginfo] target(s) in 0.05s
    Running `target/debug/orchestrator worker --name recheck`
Worker successfully registered: 200 OK
Status::Online...[200 OK]
Status::Online...[200 OK]
Status::Online...[200 OK]
Status::Online...[200 OK]
Status::Online...[200 OK]
Status::Online...[200 OK]
Status::Online...[200 OK]
Status::Online...[200 OK]
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Status::Online...[200 OK]
Status::Online...[200 OK]
Status::Online...[200 OK]
```

manish-ach@macs-MacBook-Air:~/code/CI-CD-orchestrator SCROLL: 0/23

```
manish-ach@macs in repo: CI-CD-orchestrator on ʘ main [!+] is ʘ v0.1.0 via ʘ v1.90.0 t
ook 0s
└─λ curl -s 127.0.0.1:8080/api/workers | jq
[
  {
    "name": "recheck",
    "status": "online",
    "last_heartbeat": "2026-07-02T20:58:14.044261+05:45",
    "job_id": null
  }
]

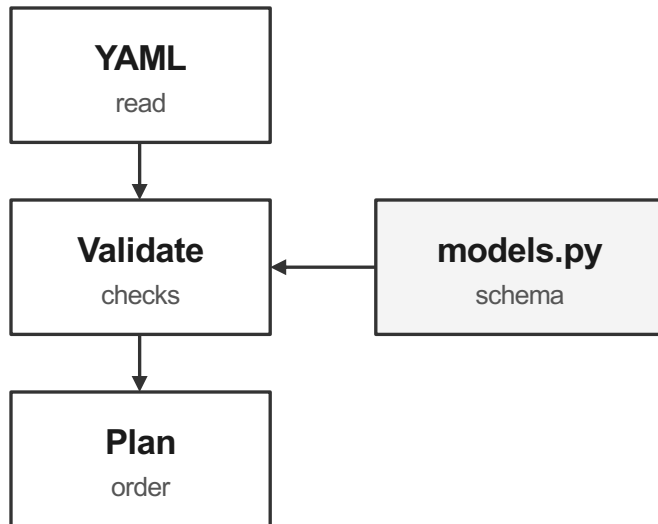
manish-ach@macs in repo: CI-CD-orchestrator on ʘ main [!+] is ʘ v0.1.0 via ʘ v1.90.0 t
ook 0s
└─λ
```

Module — Pipeline Parser



Turns a YAML file into an ordered job plan

- Reads pipeline.yml into typed objects
- Validates: unknown stages, bad references, cycles
- Topological sort -> execution order
- CLI: validate and plan commands



Done

Next: emit full job details (command, image) to the coordinator


```
└─ python3 cli.py validate bad_stage.yml
Reading 'bad_stage.yml' ...
```

Pipeline: invalid-stage

```
Stages (1): build
Jobs (1):
- test [stage: deploy]
```

Running checks:

[ok] schema and required fields

✗ Validation error: Job 'test' uses unknown stage 'deploy'

```
└─ python3 cli.py validate pipeline.yaml
Reading 'pipeline.yaml' ...
```

Pipeline: demo

```
Stages (2): build, test
Jobs (2):
- compile [stage: build]
- unit-tests [stage: test] needs: compile
```

Running checks:

[ok] schema and required fields
[ok] stages exist for every job
[ok] all dependencies resolve
[ok] no dependency cycles

✓ Pipeline 'demo' is valid (2 jobs across 2 stages)

```
└─ python3 cli.py validate bad_self_dep.yml
Reading 'bad_self_dep.yml' ...
```

Pipeline: self-dependency

```
Stages (1): build
Jobs (1):
- compile [stage: build] needs: compile
```

Running checks:

[ok] schema and required fields

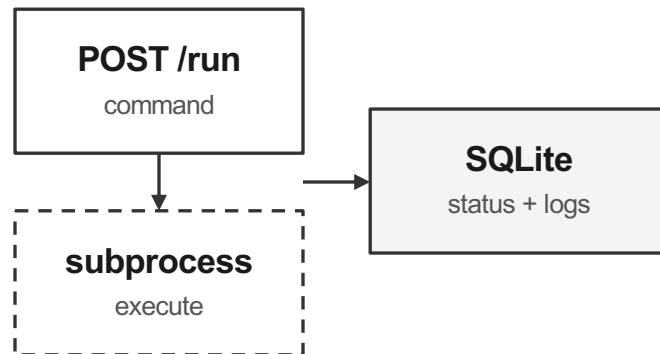
✗ Validation error: Job 'compile' cannot depend on itself

Module — Command Executor



Runs a job in a container and reports the result

- FastAPI service; the worker calls it over HTTP
- Runs the command as a subprocess (Docker next)
- Captures logs, exit code, timeout, cancel
- Stores every job in SQLite (survives restart)



In progress

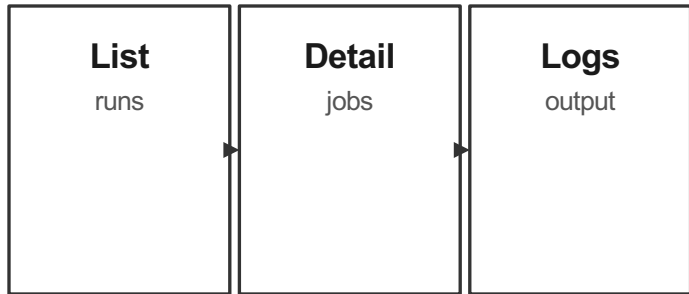
Foundation (config, DB schema, API) built; executor is the current work

Module — Dashboard



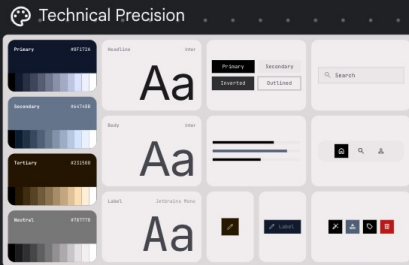
Live view of pipelines, jobs and logs

- Three pages: run list, run detail, job logs
- Colour-coded status (green / red / yellow / grey)
- Auto-refresh every 3 seconds (plain fetch)
- Pure HTML / CSS / JS — no framework needed



In progress

UI shell built; data layer (fetch + render) is next



Pipelines - Orchestrator

Orchestrator Build v2.4.0

Pipelines Settings

Pipeline Dashboard **prod/real-cluster** Search pipelines...

TOTAL PIPELINES 42 HEALTH SCORE 98.2% ACTIVE RUNS 3

Recent Pipeline Runs

NAME	BRANCH	STATUS	DURATION	STARTED AT
auth-service-prod	!-main	PAUSED	126s	2 mins ago
payment-gateway	!-feat/stripe-v1	RUNNING	45s	Just now
data-analyzer-core	!-main	FAILED	18s	10 mins ago
frontend-web-app	!-main	PAUSED	210s	1 hour ago
inventory-svc	!-fix/mem-leak	PAUSED	89s	2 hours ago
notification-worker	!-main	PAUSED	45s	3 hours ago
search-indexer	!-refactor/elastic	RUNNING	312s	5 mins ago
api-documentation	!-main	PAUSED	12s	Yesterday
cache-manager	!-main	FAILED	5s	Yesterday
user-profile-svc	!-develop	RUNNING	182s	Just now

Showing 12 of 42 pipelines

Alex Chen Platform Eng

Pipeline Details - Orchestrator

Orchestrator Build v2.4.0

Pipelines Settings

Pipeline Dashboard **prod/real-cluster** Search logs...

pipeline-PL-99231 RUNNING

PIPELINE STAGES

- Build PASSED
- Unit Tests RUNNING
- Security Scan PASSED
- Deploy Production FAILED

LOG UNIT TESTS

```
1 2023-10-24 14:02:11 [INFO] Starting test suite runner ...
2 2023-10-24 14:02:11 [INFO] Found 101 test files to execute.
3 2023-10-24 14:02:11 [INFO] Scheduling: test_auth.service.spec.ts
4 2023-10-24 14:02:11 [INFO] Scheduling: test_auth.service.spec.ts (1/26)
5 2023-10-24 14:02:11 [INFO] Scheduling: test_auth.service.spec.ts
6 2023-10-24 14:02:11 [INFO] Scheduling: test_auth.service.spec.ts
7 2023-10-24 14:02:11 [INFO] Scheduling: test_auth.service.spec.ts
8 2023-10-24 14:02:11 [INFO] Scheduling: test_auth.service.spec.ts
9 2023-10-24 14:02:11 [INFO] Scheduling: test_auth.service.spec.ts
10 2023-10-24 14:02:11 [INFO] Scheduling: test_auth.service.spec.ts
11 2023-10-24 14:02:11 [INFO] Scheduling: test_auth.service.spec.ts
12 2023-10-24 14:02:11 [INFO] Scheduling: test_auth.service.spec.ts
13 2023-10-24 14:02:11 [INFO] Scheduling: test_auth.service.spec.ts
14 2023-10-24 14:02:11 [INFO] Scheduling: test_auth.service.spec.ts
15 2023-10-24 14:02:11 [INFO] Scheduling: test_auth.service.spec.ts
16 2023-10-24 14:02:11 [INFO] Scheduling: test_auth.service.spec.ts
17 2023-10-24 14:02:11 [INFO] Scheduling: test_auth.service.spec.ts
18 2023-10-24 14:02:11 [INFO] Scheduling: test_auth.service.spec.ts
19 2023-10-24 14:02:11 [INFO] Scheduling: test_auth.service.spec.ts
20 2023-10-24 14:02:11 [INFO] Scheduling: test_auth.service.spec.ts
21 2023-10-24 14:02:11 [INFO] Scheduling: test_auth.service.spec.ts
22 2023-10-24 14:02:11 [INFO] Scheduling: test_auth.service.spec.ts
23 2023-10-24 14:02:11 [INFO] Scheduling: test_auth.service.spec.ts
24 2023-10-24 14:02:11 [INFO] Scheduling: test_auth.service.spec.ts
25 2023-10-24 14:02:11 [INFO] Scheduling: test_auth.service.spec.ts
26 2023-10-24 14:02:11 [INFO] Scheduling: test_auth.service.spec.ts
27 2023-10-24 14:02:11 [INFO] Scheduling: test_auth.service.spec.ts
```

ADMIN_USER admin@orchestrator.com

Job Logs - Orchestrator

Orchestrator Build v2.4.0

Pipelines Settings

Pipeline Dashboard **prod/real-cluster** Search logs...

deploy-production-env RUNNING

START TIME Oct 24, 2023 - 14:02:44 DURATION 00m 45s NUMBER aws-us-east-1-large-02

```
1 [INFO] Initializing build environment...
2 [INFO] Retrieving source code from repository: main branch
3 [INFO] Repository cloned successfully. Commit ID: 7f8a1b2c
4 [WARN] Increased library detected: 'old-version-v1'. Performance might be affected.
5 [INFO] Starting dependency installation...
6 [INFO] Uploading build artifacts to S3 bucket...
7 [WARN] Running integration tests on module 'auth'...
8 [INFO] Running integration tests on module 'auth'...
9 [WARN] Verifying S3 certificate validity...
10 [INFO] Verifying S3 certificate validity...
11 [WARN] Processing container layer: dfa1...
12 [INFO] Processing container layer: dfa1...
13 [WARN] Optimizing asset delivery with CDN invalidation...
14 [INFO] Optimizing asset delivery with CDN invalidation...
15 [WARN] Syncing database migrations...
16 [INFO] Syncing database migrations...
17 [WARN] Optimizing asset delivery with CDN invalidation...
18 [INFO] Optimizing asset delivery with CDN invalidation...
19 [WARN] Uploading build artifacts to S3 bucket...
20 [INFO] Uploading build artifacts to S3 bucket...
21 [WARN] Verifying S3 certificate validity...
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25 [WARN] Syncing database migrations...
26 [INFO] Syncing database migrations...
27 [WARN] Optimizing asset delivery with CDN invalidation...
28 [INFO] Optimizing asset delivery with CDN invalidation...
29 [WARN] Uploading build artifacts to S3 bucket...
30 [INFO] Uploading build artifacts to S3 bucket...
31 [WARN] Running integration tests on module 'auth'...
32 [INFO] Running integration tests on module 'auth'...
33 [WARN] Syncing database migrations...
34 [INFO] Syncing database migrations...
35 [WARN] Processing container layer: dfa1...
36 [INFO] Processing container layer: dfa1...
37 [WARN] Health check passed for instance i-dfa1...
38 [INFO] Health check passed for instance i-dfa1...
39 [WARN] Running integration tests on module 'auth'...
40 [INFO] Running integration tests on module 'auth'...
41 [WARN] Syncing database migrations...
42 [INFO] Syncing database migrations...
43 [WARN] Processing container layer: dfa1...
44 [INFO] Processing container layer: dfa1...
45 [WARN] Health check passed for instance i-dfa1...
46 [INFO] Health check passed for instance i-dfa1...
```

ADMIN_USER admin@orchestrator.com

Progress across modules

Coordinator (Rust)

Done

API, heartbeats, failure detection, claim/report

Pipeline Parser (Python)

Done

Parser, validator, planner, CLI, tests

Command Executor (FastAPI)

In progress

Config + DB + app skeleton; executor next

Dashboard (HTML/JS)

In progress

UI shell for 3 pages; data layer next

What works end-to-end today

One pipeline run, step by step



Kill a worker mid-job and the coordinator detects the failure within seconds.

Next step: run each job inside a Docker container instead of a bare subprocess.

What's to be done

Coordinator	Command Executor	Dashboard	Integration
• Redis queue + PostgreSQL • Requeue dead job + stage order • Run jobs in Docker	• Subprocess executor + logs • Timeout & cancel • Finish endpoints	• Fetch + 3s refresh • Render live data • Cross-origin (CORS)	• Wire modules over HTTP • docker-compose stack • End-to-end tests

Roadmap to final

1

Complete execution

Executor + coordinator job requeue & ordering

2

Connect & display

Dashboard data layer; wire the modules together

3

Containerize

Run jobs in Docker; docker-compose the stack

4

Harden

End-to-end tests, evaluation, documentation

Summary

- **A distributed, containerized CI/CD platform — four independent modules.**
- Done: the Rust coordinator and the pipeline parser.
- In progress: the command executor and the dashboard data layer.
- To do: Docker execution, integration, end-to-end testing.

The distributed core works today; remaining work is execution, integration and containerization.

Thank you · Questions?